

Training Configuration and Summary:

Sequencing:

- timesteps: 60
- batch_size: 128

Model Hyperparameters:

- min_units: 128
- max_units: 256
- units_step: 64
- dropout_min: 0.2
- dropout_max: 0.3
- dropout_step: 0.1
- activation_functions: ['tanh']
- optimizers: ['adam', 'Nadam']

Tuning:

- max_epochs: 30
- hyperband_iterations: 3

Directories:

- directory: my_dir
- project_name: bilstm_hyperparameter_tuning

BiLSTM Training Report

Report Generated: 2025-04-09_22-01-42

Training Start: 2025-04-09 20:42:09

Training End: 2025-04-09 21:54:46

Training Duration: 4356.66 seconds

Best Hyperparameters:

Number of Layers: 2

Activation: tanh

Optimizer: Nadam

Dropout Rate: 0.20

Units per Layer:

Layer 1: 256 units, Layer 2: 128 units

Performance Metrics:

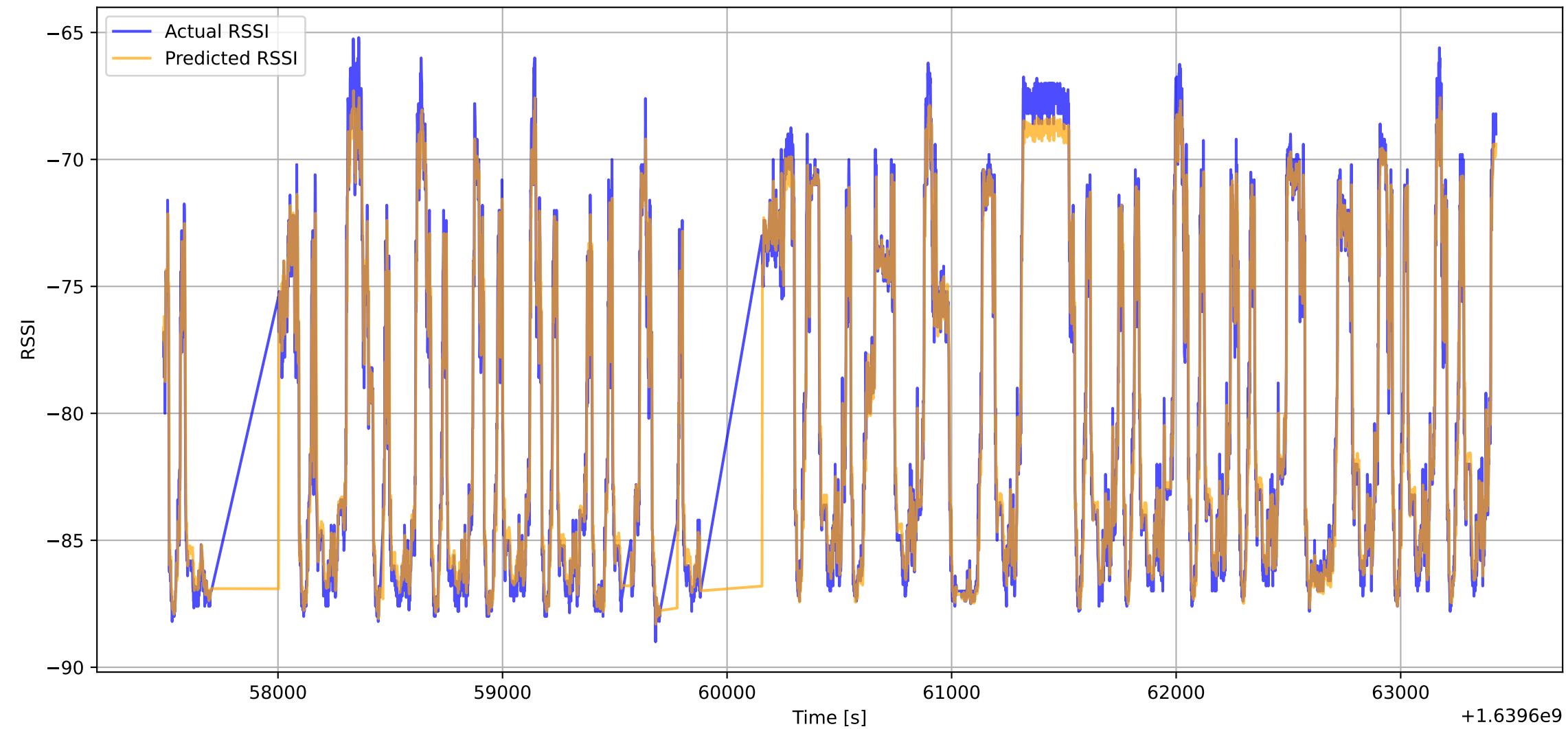
MAE: 0.9226

MSE: 1.7965

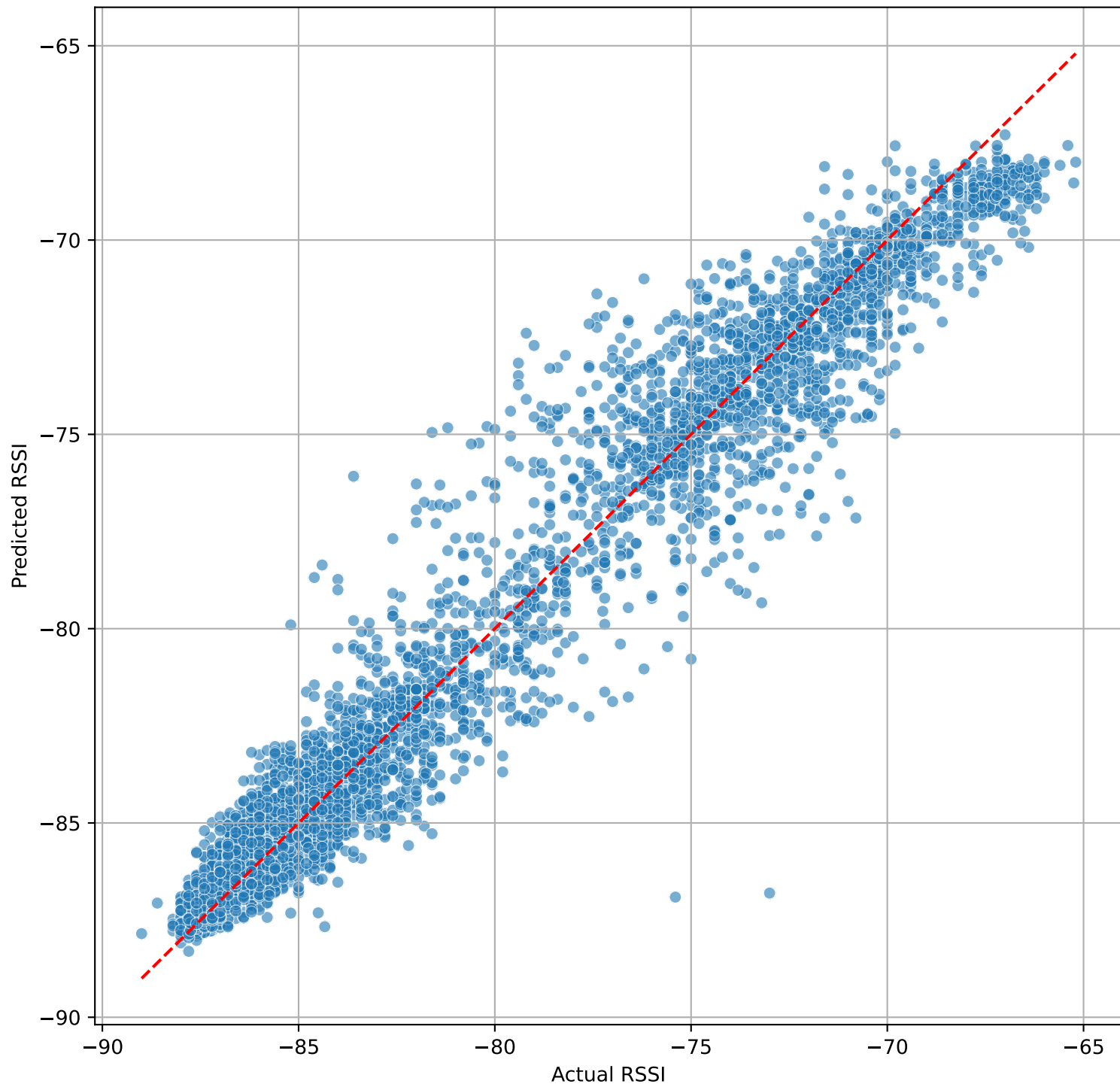
RMSE: 1.3403

R2 Score: 0.9592

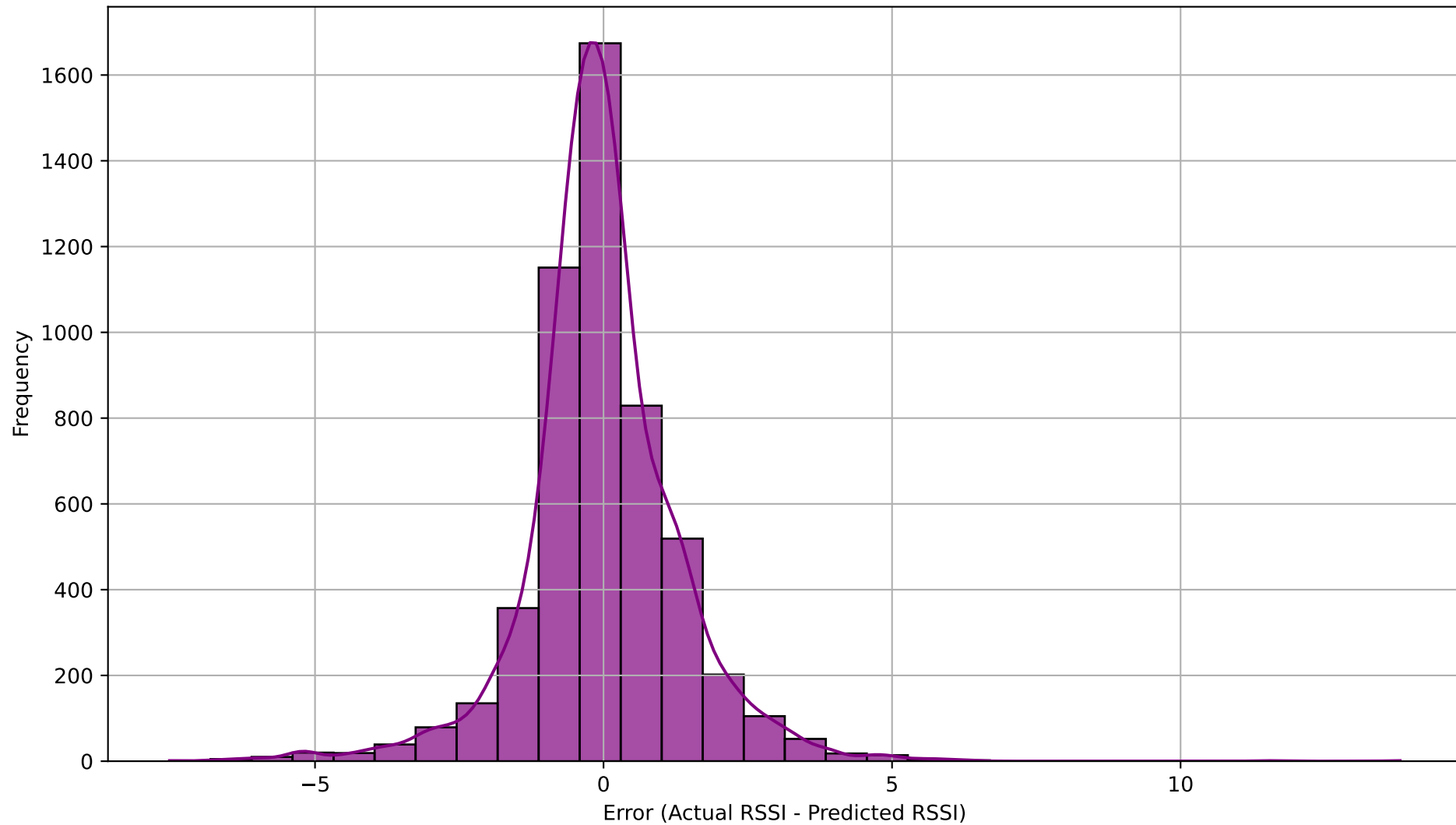
Actual vs Predicted RSSI Over Time



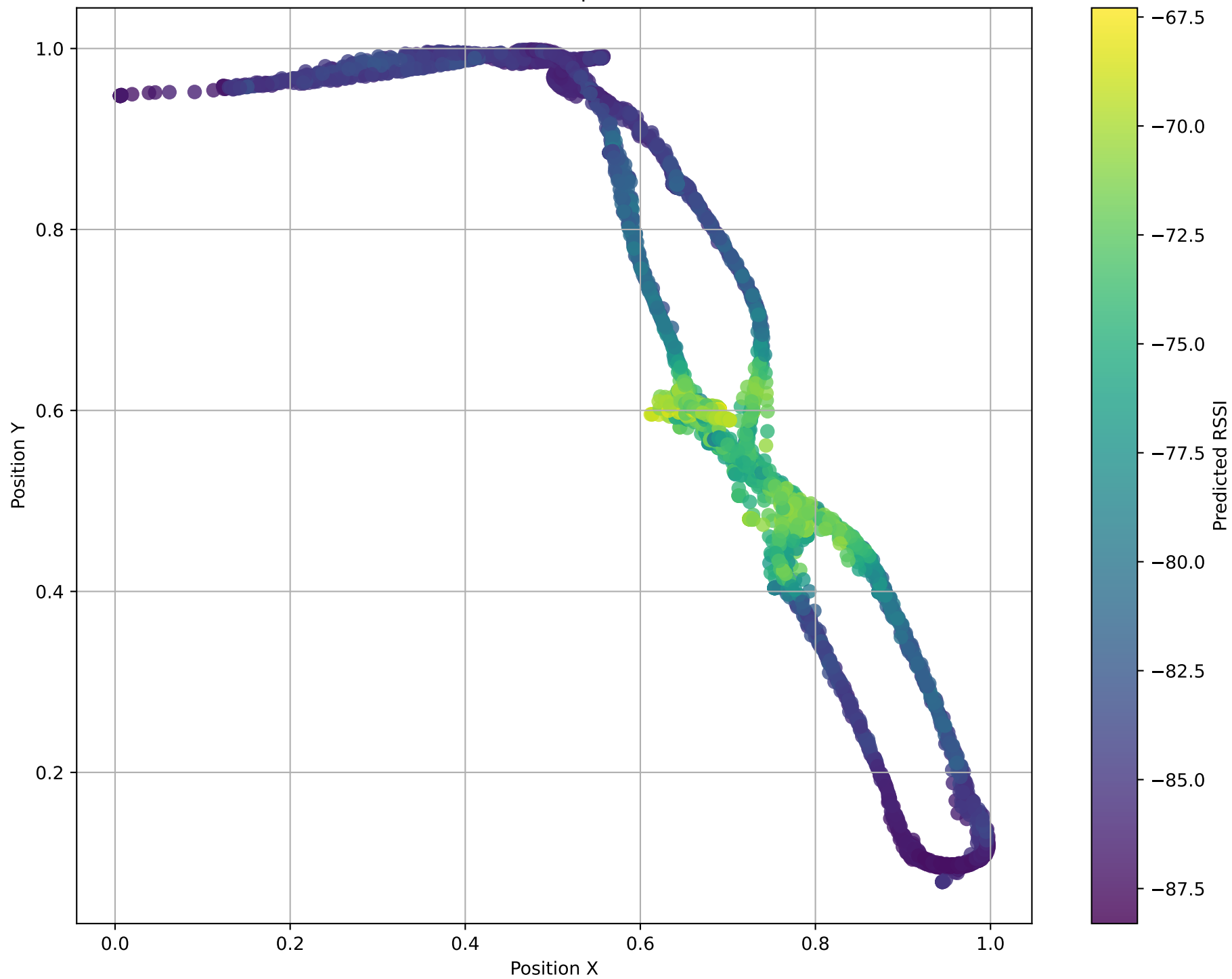
Actual vs Predicted RSSI



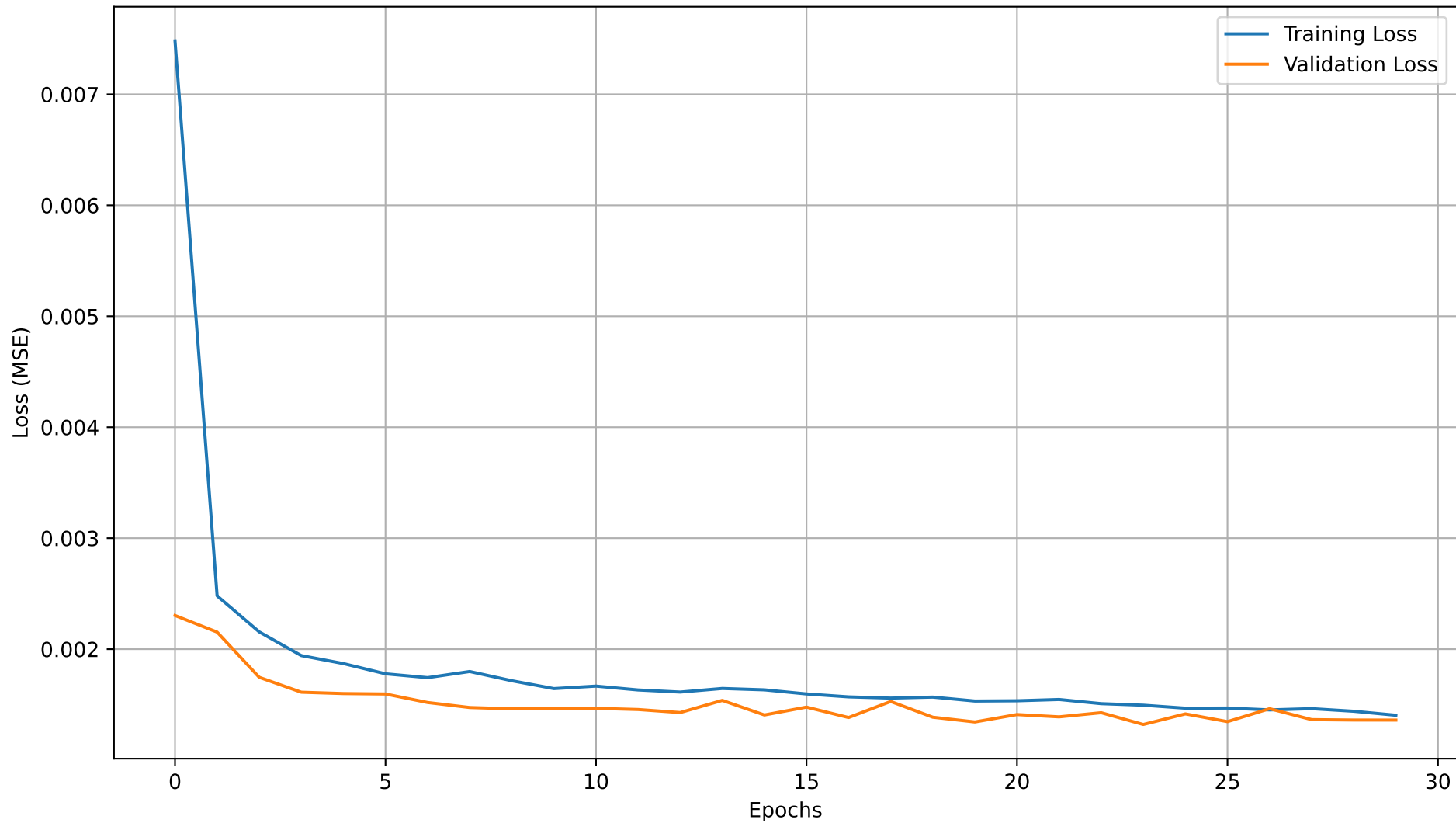
Distribution of Prediction Errors



Predicted RSSI in Spatial Context



Training and Validation Loss Over Epochs



Training and Validation MAE Over Epochs

